

Inferential Statistics



① Inferential Statistics : Inferential Statistics refers to methods that rely on probability theory and Distributions to Predict Population Values based on Sample Data.

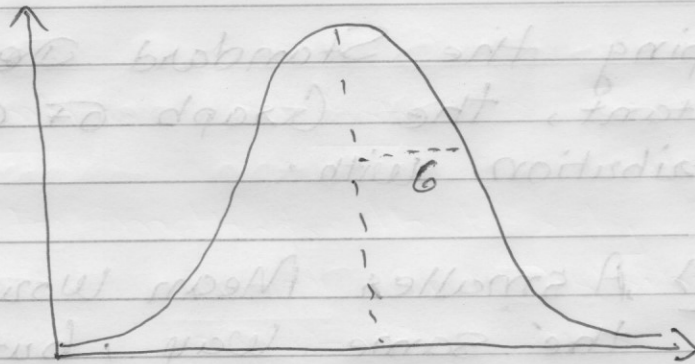
② Distribution :

In Statistics, when we use the term Distribution, we usually mean a Probability Distribution.

i.e. Normal Distribution,
Binomial Distribution,
Uniform Distribution.

A Distribution is a Function that shows the possible values for a variable and how often they occur.

Each Probability Distribution has a Visual Representation. It is a graph describing the likelihood of occurrence of every event.



mean =
mode =
median

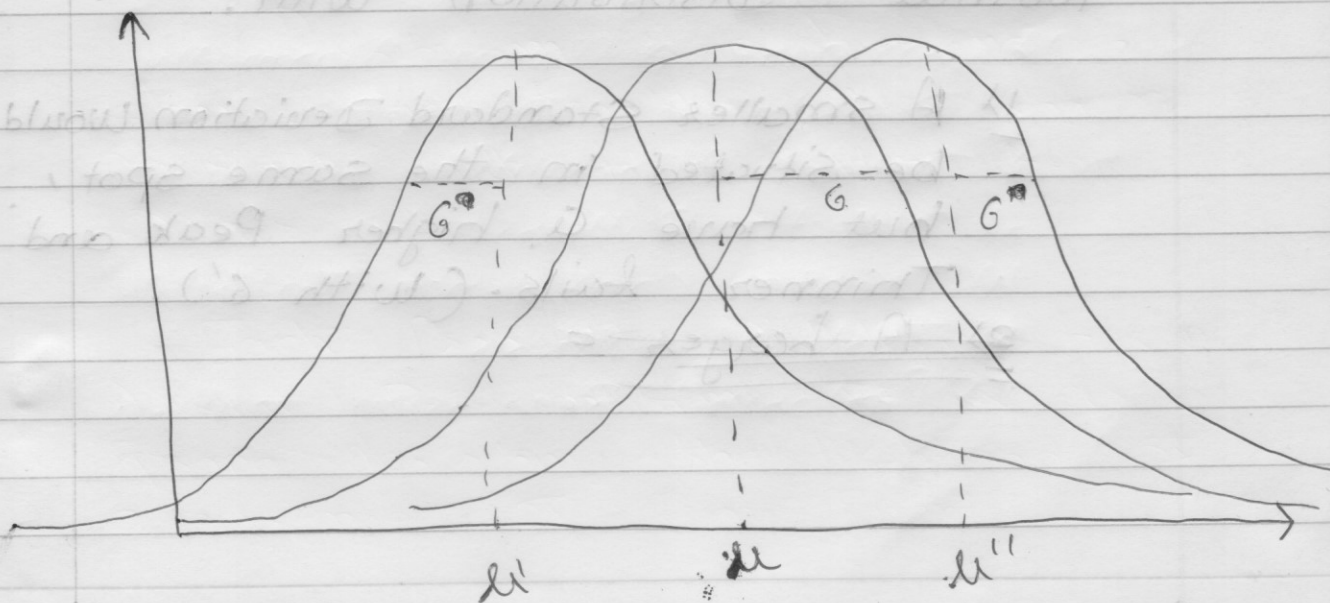
$N \sim (μ, σ^2)$; N = Normal

$\sim$ = Distribution

$μ$ = mean

$σ^2$ = Variance.

② Controlling For Standard Deviation:





keeping the Standard Deviation Constant, the Graph of a Normal Distribution With:

1) A smaller Mean would look in the same way, but be situated to the left.

(Curve with mean = μ')

2) A Larger mean would look in the same way, but be situated to the right.

(Curve with mean = μ'')

② Controlling For the Mean:

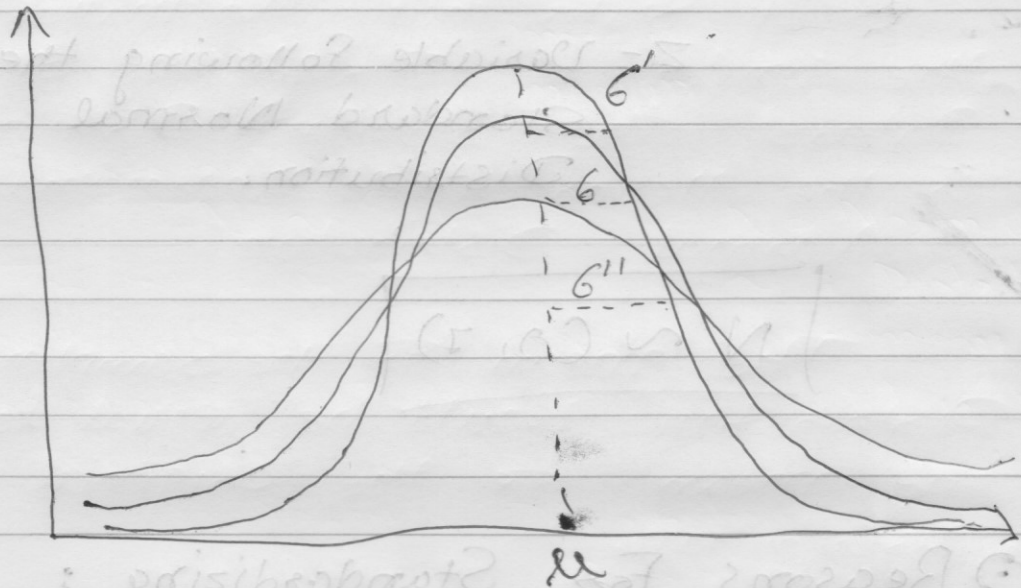
keeping the mean constant, a Normal Distribution with:

1) A smaller Standard Deviation would be situated in the same spot, but have a higher Peak and Thinner tails. (with σ').

2) A larger ~~Standard Deviation~~



2) A Larger Standard Deviation would be situated in the same ~~place~~ Spot, but have a lower peak and Fatter tails. (with $6''$).



⇒ The Standard Normal Distribution:

The Standard Normal Distribution is a Particular case of The Normal Distribution with a Mean of 0 and a Standard Deviation of 1.



Every Normal Distribution can be Standardized using the Standardization Formula:

$$Z = \frac{x - \mu}{\sigma}$$

Z = Variable following the Standard Normal Distribution.

$$N \sim (\mu, \sigma^2)$$

Reasons For Standardizing :

It allows us to :

- 1) Compare different Normally Distributed Datasets.
- 2) Detect Normality.
- 3) Detect Outliers.
- 4) Create Confidence Intervals.
- 5) Test Hypothesis.
- 6) Perform Regression Analysis.



Library

① Way to Standardize :

We want to Transform From

$$N \sim (\mu, \sigma^2)$$

to $N \sim (0, 1)$

↳ Subtracting Mean from all observations would cause a transformation from $N \sim (\mu, \sigma^2)$ to $N \sim (0, \sigma^2)$.

↳ Dividing all observations by Standard Deviation would cause a transformation from $N \sim (0, \sigma^2)$ to $N \sim (0, 1)$.



The Central Limit

Theorem

The Central Limit Theorem (CLT) is one of the greatest Statistical Insights.

It states that, "No matter the underlying Distribution of the Dataset, the Sampling Distribution of the Means would approximate a Normal Distribution".

The mean of the Sampling Distribution would be equal to the mean of the original Distribution.

The Variance of the Sampling Distribution would be 'n' times smaller than the variance of the original Distribution. (n = size of the Sample).



i.e. No matter the Underlying Distribution:

↳ Distribution of $\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4, \dots, \bar{x}_k$ would be

$$N \sim \left(\mu, \frac{\sigma^2}{n} \right); n = \text{Size of Samples.}$$

↳ The more Samples, the closer to Normal will be Sampling Distribution of means. ($k \rightarrow \infty$)

↳ The bigger the Samples, the closer to Normal will be Sampling Distribution of means. ($n \rightarrow \infty$)

⊙ Use of CLT:

The CLT allows us to assume Normality for many different Variables.



That is very useful for
Confidence Intervals, Hypothesis
Testing and Regression Analysis.

A distribution of the sample means will be

$$N \sim \left(\mu, \frac{\sigma^2}{n} \right) \quad \text{if } n \text{ is large}$$

if the above samples, the
closer to Normal will be
sampling distribution of
means. ($n \rightarrow \infty$)

The bigger the samples,
the closer to Normal will
be sampling distribution
of means.
($n \rightarrow \infty$)

③ The CLT:

The CLT allows us to
assume Normality for many
different variables.

Standard Error



PAGE NO. 56

DATE

The Standard Error is the Standard Deviation of the Distribution formed by the Sample means.

i.e. Standard Deviation of the Sampling Distribution.

$$\text{Variance of Sampling Distribution of mean} = \frac{\sigma^2}{n}$$

$$\therefore \text{Standard Error} = \frac{\sigma}{\sqrt{n}}$$

It Describes Variability of ~~Diff~~ Means of Different Samples we extracted.

Estimators and Estimates

=> Estimators :

Broadly, an Estimator is a Mathematical Function that Approximates a Population Parameter depending only on Sample information.

e.g.

Term	Estimator	Parameter
Mean	$\bar{x}$	μ
Variance	s^2	σ^2
Correlation	r	ρ

↳ Properties of Estimators :

1) Bias : The expected value of an Unbiased Estimator is the Population Parameter.

The bias in this case is 0.

If the



If the expected value of an estimator is $(\text{parameter} + b)$, then the bias is b .

(2) Efficiency: The most efficient Estimator is the one with the Smallest Variance.

$\Rightarrow$ Estimates:

An Estimate is the output we get from the Estimator when we apply the Formula.

There are two types of Estimates:

1) Point Estimates:

It is a single value.

e.g. 7, 5, 0.32, 555.97.

(2) Confidence Intervals:

It is an Interval.

e.g. $(7, 5)$, $(12, 33)$, $(-0.5, 0.95)$



Confidence Intervals are much more Precise than Point Estimates.

Confidence Intervals are preferred over Point Estimates while making inferences.